

EXCEPTION HANDLING

Q1 Handling ArithmeticException

10 marks

Write a Java program to divide two integers entered by the user. If the denominator is zero, handle the exception using a try-catch block and display an appropriate error message.

Your Code

```
1 public class arithmeticException{
2     public static void main(String[] args){
3         int a=0;
4         int b=0;
5         try{
6             int result = a/b;
7             System.out.println("Result="+result);
8         }
9         catch(ArithmeticException e){
10             System.out.println("error: cannot divide by zero.");
11         }
12     }
13 }
```

Run

Save

Input

stdin input

Output

error: cannot divide by zero.

Q2 Handling ArrayIndexOutOfBoundsException

10 marks

Write a Java program to create an integer array of five elements. Ask the user to enter an index and display the corresponding element. If the index is outside the valid range, handle the exception and display an error message.

Your Code

```
1 public class OutOfBounds{
2     public static void main(String[] args){
3         int []arr={10,20,30,49};
4         int index=6;
5         try{
6             System.out.println("element="+arr[index]);
7         }
8         catch(ArrayIndexOutOfBoundsException e){
9             System.out.println("error:invalid index");
10        }
11    }
12 }
```

Run

Save

Input

stdin input

Output

error:invalid index

Q3 Handling NumberFormatException

10 marks

Write a Java program to accept a number as a string and convert it into an integer using `Integer.parseInt()`. If the input is not a valid integer, handle the exception and display an appropriate message.

Your Code

```
1 public class numberformat{
2     public static void main(String[] args) {
3         String s="ABC";
4         try{
5             int num=Integer.parseInt(s);
6             System.out.println("number="+num);
7         }
8         catch(NumberFormatException e){
9             System.out.println("error:invalid number format");
10        }
11    }
12 }
```

 Run Save**Input**

stdin input

Output

error:invalid number format

Q4 Handling Multiple Exceptions

10 marks

Write a Java program that performs the following operations:

Accept two integers from the user.

Divide the first number by the second.

Store the result in an array.

Handle both `ArithmeticException` and `ArrayIndexOutOfBoundsException` using multiple catch blocks.

Your Code

```
1 public class Main{
2     public static void main(String[] args) {
3         int a=10;
4         int b=0;
5         int[] arr=new int[2];
6         try{
7             int result=a/b;
8             arr[5]=result;
9         }
10        catch(ArithmeticException e){
11            System.out.println("error:cannot divide by zero");
12        }
13        catch(ArrayIndexOutOfBoundsException e){
14            System.out.println("error:invalid array index");
15        }
16    }
17 }
```

Input

stdin input

Output

error:cannot divide by zero